MEMORY HIERARCHY

Organization of Memory in Computer Systems | DLCO Unit IV

Digital Logic and Computer Organization (DLCO) | Unit IV
JNTUK R23 Regulation | Department of Computer Science & Engineering

Presented by (Team of 5 Members - CAI 2-1):
- Member 1: Karthikeya (Intro & Hierarchy Structure)
- Member 2: Nikhil (Semiconductor RAM & ROM)
- Member 3: Teja (Cache Memory & Performance)
- Member 4: Srinivas (Virtual Memory & Management)
- Member 5: Manohar (Secondary Storage & Summary)

Seminar Presentation & Comprehensive Architectural Study

1. Introduction to Memory Organization

Presented by: Member 1 (Karthikeya) | Basic Concepts & Trade-offs

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Role of Memory in Systems:

- Memory stores programs, instructions, operational data, and computation results.
- Crucial subsystem directly determining processor utilization and throughput.
- CPU executes instructions at nanosecond clock speeds requiring rapid data access.
- A single memory technology cannot satisfy both maximum speed and maximum capacity.

The Core Trade-offs & Hierarchy Strategy:

- Speed vs. Cost: Faster memory technologies are substantially more expensive per bit.
- Capacity vs. Speed: Larger capacity memories exhibit longer physical access latencies.
- Hierarchy Strategy: Combines multiple memory technologies in a tiered structure.
- Goal: Delivers nearly the speed of fastest memory with the capacity and cost of cheapest.

2. Structure of Memory Hierarchy

Presented by: Member 1 (Karthikeya) | The Memory Pyramid

The Memory Pyramid (Top to Bottom):

1. CPU Internal Registers : Access < 1 ns | Bytes to KB | Highest cost per bit
2. Cache Memory (L1, L2, L3) : Fast SRAM bridging processor & main memory speeds
3. Main Memory (RAM) : System DRAM holding active applications and datasets (50-100 ns)
4. USB / Flash Memory : High-density solid-state intermediate storage
5. Magnetic Disk / Hard Disk : Secondary storage; high capacity, low cost per bit
6. Magnetic Tapes / Drives : Archival storage; highest capacity, lowest cost per bit

Key Hierarchy Trends:

- Downward trend: Access latency increases, capacity expands, cost per bit decreases.

3. Semiconductor RAM Memories

Presented by: Member 2 (Nikhil) | Random Access Memory (SRAM & DRAM)

Static RAM (SRAM):

- Storage: Built using bistable flip-flop logic circuits (4 to 6 transistors per cell).
- Speed: Very fast access times (sub-nanosecond to few nanoseconds).
- Refreshing: Does NOT require dynamic periodic refreshing while power is maintained.
- Primary Application: Used exclusively for CPU internal and external Cache memory.

Dynamic RAM (DRAM):

- Storage: Stores bits as electrical charges in tiny capacitor-transistor pairs.
- Refreshing: Charge leaks away continuously; requires periodic refreshing cycles.
- Density & Cost: Extremely high bit density, simple cell layout, very low cost per bit.
- Primary Application: Standard choice for computer Main Memory (System RAM).

4. Read-Only Memory (ROM)

Presented by: Member 2 (Nikhil) | Non-Volatile Semiconductor Storage

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ROM Characteristics & Role:

- Non-Volatile: Retains program code permanently even when electrical power is disconnected.
- System Bootstrap: Holds initialization programs (BIOS / UEFI) and startup routines.
- Embedded Firmware: Stores dedicated control routines in microcontrollers & appliances.
- Normal Operation: Standard CPU operation is strictly read-only during execution.

Evolution & Types of ROM:

- Mask ROM: Hardwired during fabrication; cannot be modified or updated.
- PROM: Shipped blank and electrically programmed once using fuse links.
- EPROM: Erased by exposing chip die to ultraviolet (UV) light.
- EEPROM: Electrically Erasable PROM; erased in-circuit via electrical signals byte-by-byte.
- Flash Memory: High-density EEPROM variation erasable in large sector blocks.

5. Cache Memory Architecture

Presented by: Member 3 (Teja) | High-Speed Buffer & Multi-Level Cache

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Principles of Cache Operation:

- Buffer Role: Small, ultra-fast memory placed directly between CPU execution units & DRAM.
- Temporal Locality: Recently accessed items are likely to be accessed again in near future.
- Spatial Locality: Items stored near recently accessed items are likely to be accessed soon.
- Transparency: Operates automatically in hardware without programmer management.

Multi-Level Cache Organization:

- L1 Cache: Integrated inside CPU core; smallest (32-64 KB), fastest (sub-nanosecond).
- L2 Cache: Dedicated per-core buffer; larger (256 KB-1 MB), slightly slower access.
- L3 Cache: Shared across all processor cores; high capacity (8-64 MB).
- Access Flow: CPU checks L1 -> L2 -> L3 -> falls back to DRAM only on cache miss.

6. Cache Performance Considerations

Presented by: Member 3 (Teja) | Metrics & Access Time Analysis

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Key Performance Metrics:

- Cache Hit: Requested memory word is found directly within cache buffer.
- Cache Miss: Requested word is not present, requiring retrieval from main memory.
- Hit Ratio (h): Fraction of memory references satisfied by cache: (Hits) / (Total Accesses).
- Miss Ratio: Complement of hit ratio, equal to (1 - Hit Ratio).
- Miss Penalty: Extra latency required to fetch the required block from DRAM into cache.

Performance Analysis & Average Memory Access Time (AMAT):

- Typical Hit Ratio: 90% - 95% hit ratio drastically lowers effective memory latency.
- Formula: AMAT = Hit Time + (Miss Rate * Miss Penalty).
- High hit rates ensure CPU rarely experiences the full DRAM access delay.

7. Virtual Memory Architecture

Presented by: Member 4 (Srinivas) | Address Translation & Paging

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Concept & Hardware Support:

- Logical Address Space: Grants running programs an illusion of large, contiguous memory.
- Overcoming Limits: Programs can run even if they exceed physical RAM size.
- Memory Management Unit (MMU): Hardware component performing fast address translation.
- Decoupling: Decouples user logical addresses from actual physical memory locations.

Paging & Page Fault Mechanism:

- Pages & Frames: Virtual memory divided into pages; physical RAM into page frames.
- Page Table: Maps process virtual pages to corresponding physical RAM frames.
- Page Fault: Triggered when an active instruction references a page not currently in RAM.
- Demand Paging: OS retrieves required page from secondary storage into free RAM frame.
- Replacement Policy: Replaces inactive pages (using LRU or FIFO) when memory is full.

8. Memory Management Requirements

Presented by: Member 4 (Srinivas) | Core Operating System Functions

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Core Hardware and OS Functions:

- Relocation: Dynamic address translation allowing programs to be loaded anywhere in RAM.
- Protection: Prevents processes from unauthorized access to other memory areas.
- Base & Limit Registers: Enforce hardware protection boundaries.
- Sharing: Allows cooperating processes to access common code safely (shared DLLs).
- Logical Organization: Structures memory into logical modules, procedures, and arrays.
- Physical Organization: Coordinates transparent flow between volatile RAM & secondary disk.
- Maximizes multitasking throughput and system stability.

9. Secondary Storage Systems

Presented by: Member 5 (Manohar) | Non-Volatile Mass Storage

Role in Memory Hierarchy:

- Permanent Repository: Non-volatile storage for OS kernel, applications, and user files.
- Massive Capacity: Provides hundreds of gigabytes to several terabytes of storage volume.
- Lowest Cost per Bit: Highly economical compared to semiconductor SRAM and DRAM.
- Virtual Memory Backing: Serves as swap space holding inactive pages displaced from RAM.

Storage Media Technologies:

- Solid State Drives (SSD): NAND flash memory; no moving parts, high IOPS, fast random access.
- Hard Disk Drives (HDD): Magnetic platters with mechanical arms; high capacity, economical.
- Removable & Optical Media: USB flash drives and optical discs (CD/DVD/Blu-ray).

10. Comparison of Memory Levels

Presented by: Member 5 (Manohar) | Quantitative Summary Across Hierarchy

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Quantitative Comparison Matrix:

- CPU Registers : CMOS Flip-Flops | < 1 ns | Bytes to KB | Highest Cost/Bit
- Cache Memory : Static RAM (SRAM) | 1 - 10 ns | KB to MB | Very High Cost/Bit
- Main Memory : Dynamic RAM (DRAM) | 50 - 100 ns | 8 - 64 GB | Moderate Cost/Bit
- Secondary Storage : SSD Flash / HDD Magnetic | Micro-ms | 512GB - 4TB | Lowest Cost/Bit

Key Architectural Law on Memory Trade-offs:

- As we move downward: Access time increases by orders of magnitude,
  capacity expands dramatically, and cost per bit plummets.
- Ensures high-speed execution without prohibitive semiconductor manufacturing costs.

11. Summary & Key Takeaways

Presented by: Member 5 (Manohar) | Key Insights & Wrap-up

Seminar Synthesis:

- Unified Hierarchy: Bridges the growing processor-memory performance gap (Memory Wall).
- Complementary Roles: Fast SRAM caches mask DRAM latencies, while virtual memory pages leverage secondary storage capacity.
- Cost-Effective Performance: Delivers near-cache access speed at main memory capacity & budget.
- System Stability: OS memory management ensures isolation, protection, and relocation.

12. Conclusion & Team Acknowledgements

Department of Computer Science & Engineering | JNTUK R23

THANK YOU!

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Unit IV: Memory System Architecture & Organization